COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9-3-26	**DATE SUBMITTED:**	9-3-26
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ACTIVITY NO. 6 **Targeting Specific Nodes**

1. PROCEDURE OUTPUT

Task 1:

1.

```
  GNU nano 7.2                          site.yml
---

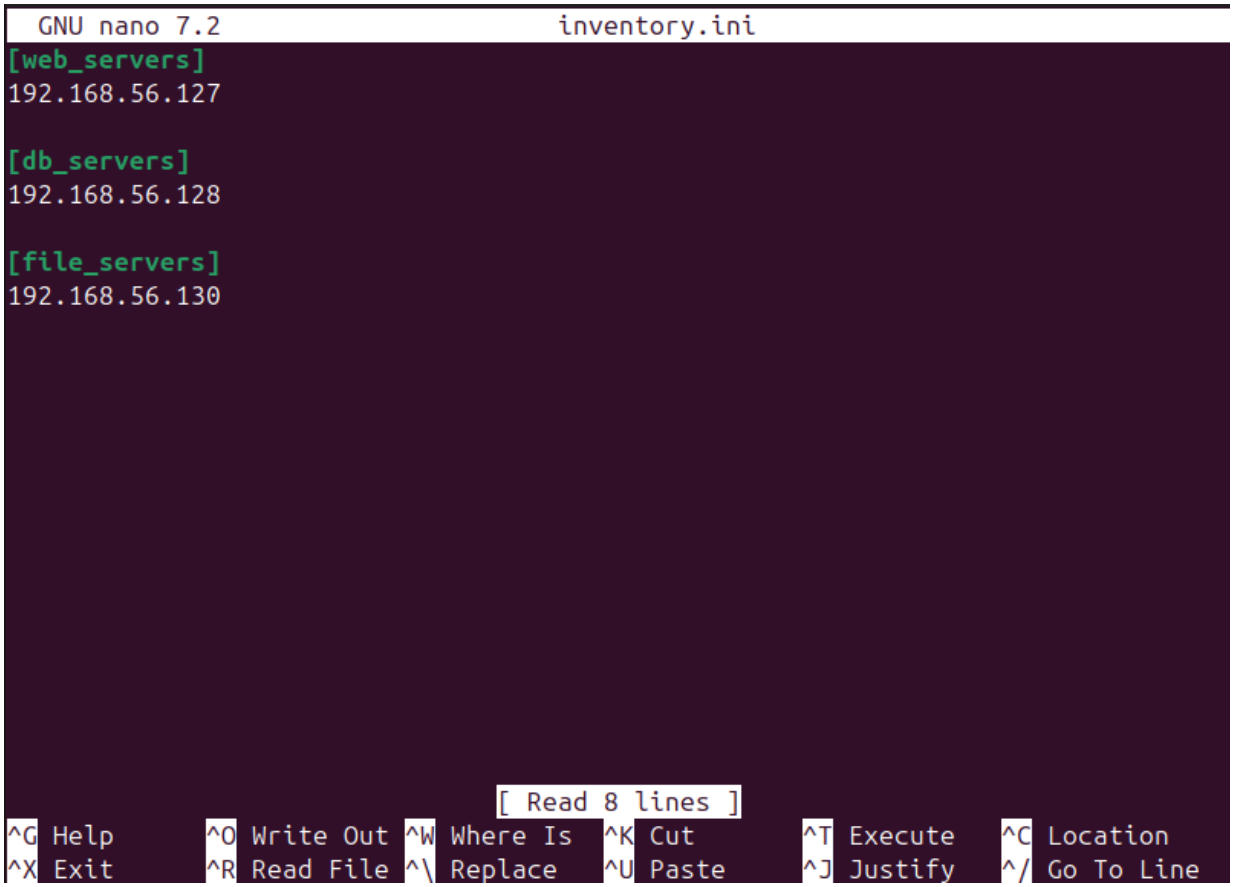
- hosts: all
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

2.


GNU nano 7.2 inventory.ini
[web_servers]
192.168.56.127
[db_servers]
192.168.56.128
[file_servers]
192.168.56.130
[Read 8 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

3.

```
  GNU nano 7.2                                site.yml *
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    apt:
      upgrade: dist
      update_cache: yes
      cache_valid_time: 3600
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

```
TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.128]
skipping: [192.168.56.129]
ok: [192.168.56.130]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.130]
ok: [192.168.56.129]
ok: [192.168.56.128]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.128]

TASK [install apache and php for Ubuntu servers] *******************************
ok: [192.168.56.128]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.128]

PLAY RECAP *********************************************************************
```

```
PLAY RECAP *********************************************************************
192.168.56.128             : ok=4    changed=0    unreachable=0    failed=0    skipped
=2    rescued=0    ignored=0
192.168.56.129             : ok=2    changed=0    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0
192.168.56.130             : ok=2    changed=0    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0
```

4.

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    services:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
TASK [install mariadb package (CentOS)] ****************************************
changed: [192.168.56.130]

TASK [Mariadb- Restarting/Enabling] ********************************************
changed: [192.168.56.130]

TASK [install mariadb package (Ubuntu)] ****************************************
skipping: [192.168.56.130]

PLAY RECAP *********************************************************************
192.168.56.128             : ok=4    changed=0    unreachable=0    failed=0    skippe
=2    rescued=0    ignored=0
192.168.56.129             : ok=2    changed=0    unreachable=0    failed=0    skippe
=1    rescued=0    ignored=0
192.168.56.130             : ok=5    changed=2    unreachable=0    failed=0    skippe
=2    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- The package was able to be installed as long as the address matches the same distribution.

5.

```
masongsong@centos:~ — systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-09-03 09:27:11 PST; 3min 13s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 8380 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, >
    Process: 8403 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.serv>
    Process: 8506 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited>
   Main PID: 8489 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 22968)
     Memory: 72.8M (peak: 100.4M)
        CPU: 1.890s
     CGroup: /system.slice/mariadb.service
             └─8489 /usr/libexec/mariadbd --basedir=/usr

Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: The second is mysql@localh>
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: you need to be the system >
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: After connecting you can s>
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: able to connect as any of >
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: See the MariaDB Knowledgeb>
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: Please report any problems>
Sep 03 09:27:10 centos mariadb-prepare-db-dir[8444]: The latest information abo>
lines 1-23
```

```
masongsong@server2:~$ systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-09-03 01:35:34 UTC; 22s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 7760 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var>
    Process: 7762 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_ST>
    Process: 7764 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && >
    Process: 7836 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_S>
    Process: 7838 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/>
   Main PID: 7824 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 12 (limit: 9531)
     Memory: 80.1M (peak: 83.2M)
        CPU: 1.555s
     CGroup: /system.slice/mariadb.service
             └─7824 /usr/sbin/mariadbd
```

6.

```
TASK [install samba package] ***************************************************
changed: [192.168.56.130]

PLAY RECAP *********************************************************************
192.168.56.128             : ok=4    changed=0    unreachable=0    failed=0    skipped
=2    rescued=0    ignored=0
192.168.56.129             : ok=5    changed=1    unreachable=0    failed=0    skipped
=2    rescued=0    ignored=0
192.168.56.130             : ok=4    changed=1    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

Task 2:

1.

```
  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

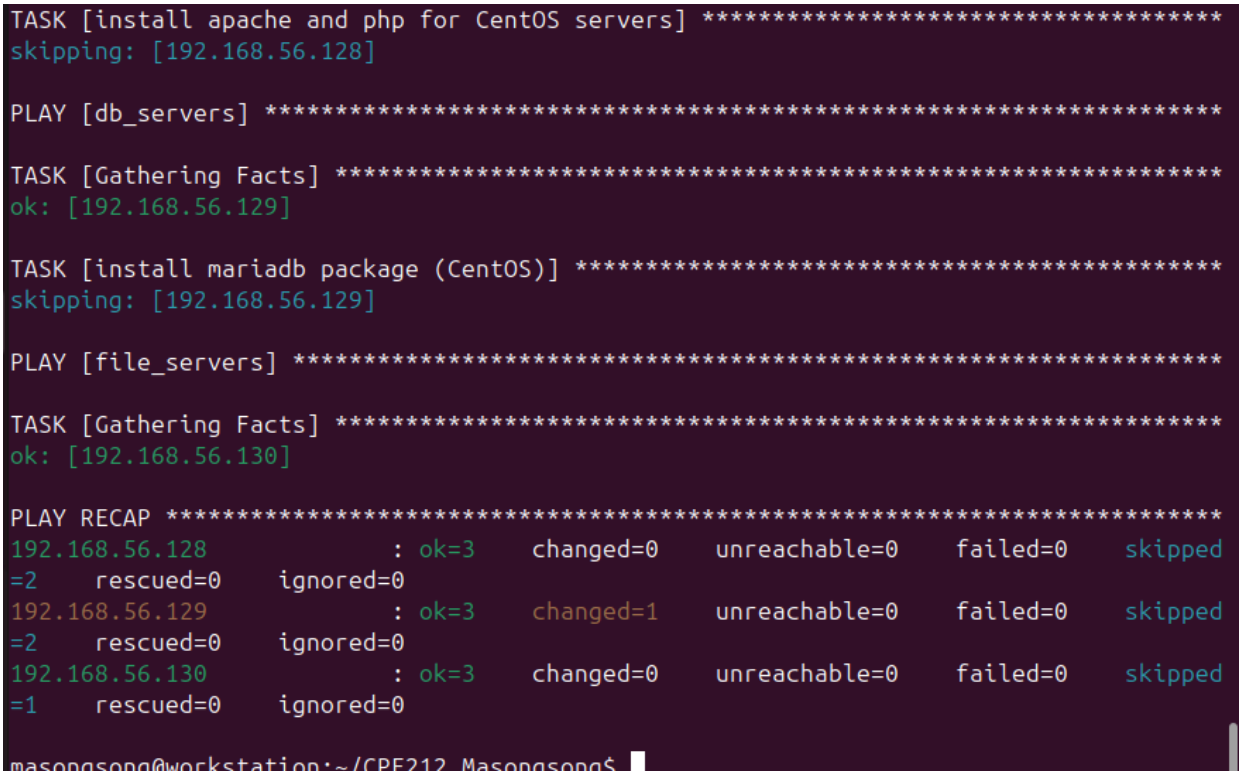
  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
masongsong@workstation:~/CPE212_Masongsong$
```

- Shows all the tags of which package/distribution is in each play in the playbook. It also plays the “always”

tag.



- Plays only the ones with the centos tag.

```
PLAY [db_servers] ***************************************************************

TASK [Gathering Facts] **********************************************************
ok: [192.168.56.129]

TASK [install mariadb package (CentOS)] *****************************************
skipping: [192.168.56.129]

TASK [install mariadb package (Ubuntu)] *****************************************
ok: [192.168.56.129]

PLAY [file_servers] *************************************************************

TASK [Gathering Facts] **********************************************************
ok: [192.168.56.130]

PLAY RECAP **********************************************************************
192.168.56.128             : ok=3    changed=0    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0
192.168.56.129             : ok=4    changed=0    unreachable=0    failed=0    skipped
=2    rescued=0    ignored=0
192.168.56.130             : ok=3    changed=0    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0

masongsong@workstation:~/CPE212_Masongsong$
```

- Plays only the ones with the “db” tag.

```
TASK [Gathering Facts] **********************************************************
ok: [192.168.56.128]

TASK [install apache and php for Ubuntu servers] ********************************
ok: [192.168.56.128]

TASK [install apache and php for CentOS servers] ********************************
skipping: [192.168.56.128]

PLAY [db_servers] ***************************************************************

TASK [Gathering Facts] **********************************************************
ok: [192.168.56.129]

PLAY [file_servers] *************************************************************

TASK [Gathering Facts] **********************************************************
ok: [192.168.56.130]

PLAY RECAP **********************************************************************
192.168.56.128             : ok=4    changed=0    unreachable=0    failed=0    skipped
=2    rescued=0    ignored=0
192.168.56.129             : ok=3    changed=0    unreachable=0    failed=0    skipped
=1    rescued=0    ignored=0
192.168.56.130             : ok=3    changed=0    unreachable=0    failed=0    skipped
```

- Plays only the ones with the “apache” tag.

```
TASK [Gathering Facts] *********************************************************
ok: [192.168.56.128]

TASK [install apache and php for Ubuntu servers] *******************************
ok: [192.168.56.128]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.128]

PLAY [db_servers] **************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.129]

TASK [install mariadb package (CentOS)] ****************************************
skipping: [192.168.56.129]

TASK [install mariadb package (Ubuntu)] ****************************************
ok: [192.168.56.129]

PLAY [file_servers] ************************************************************
```

- Plays only the ones with “apache” and “db” tags.

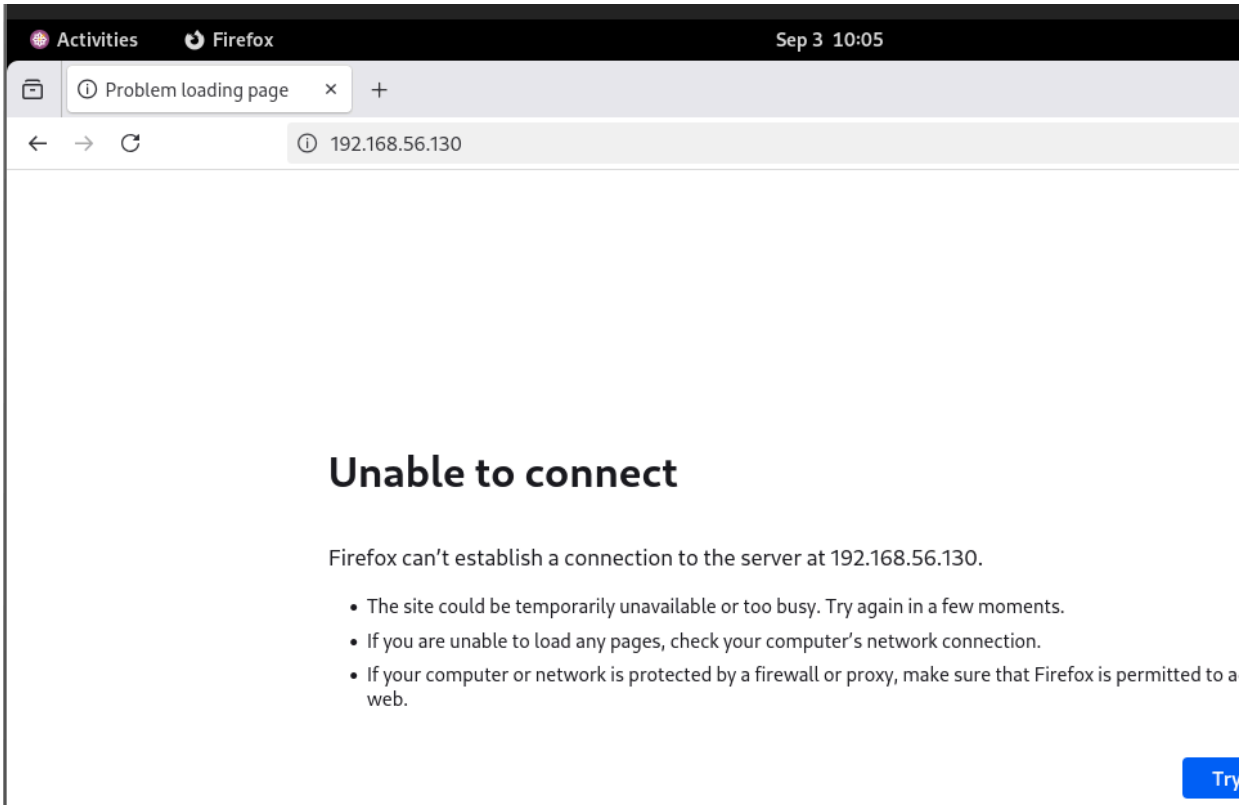
Task 3:

1.

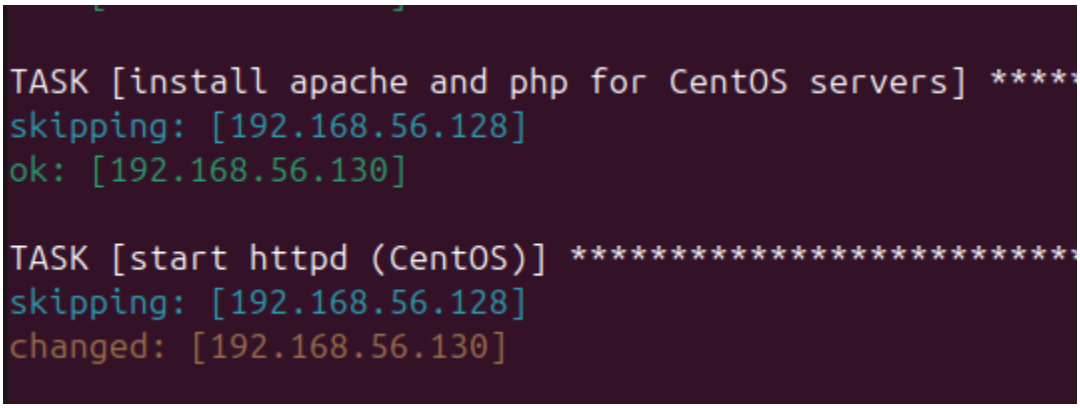
```
- name: install apache and php for CentOS servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

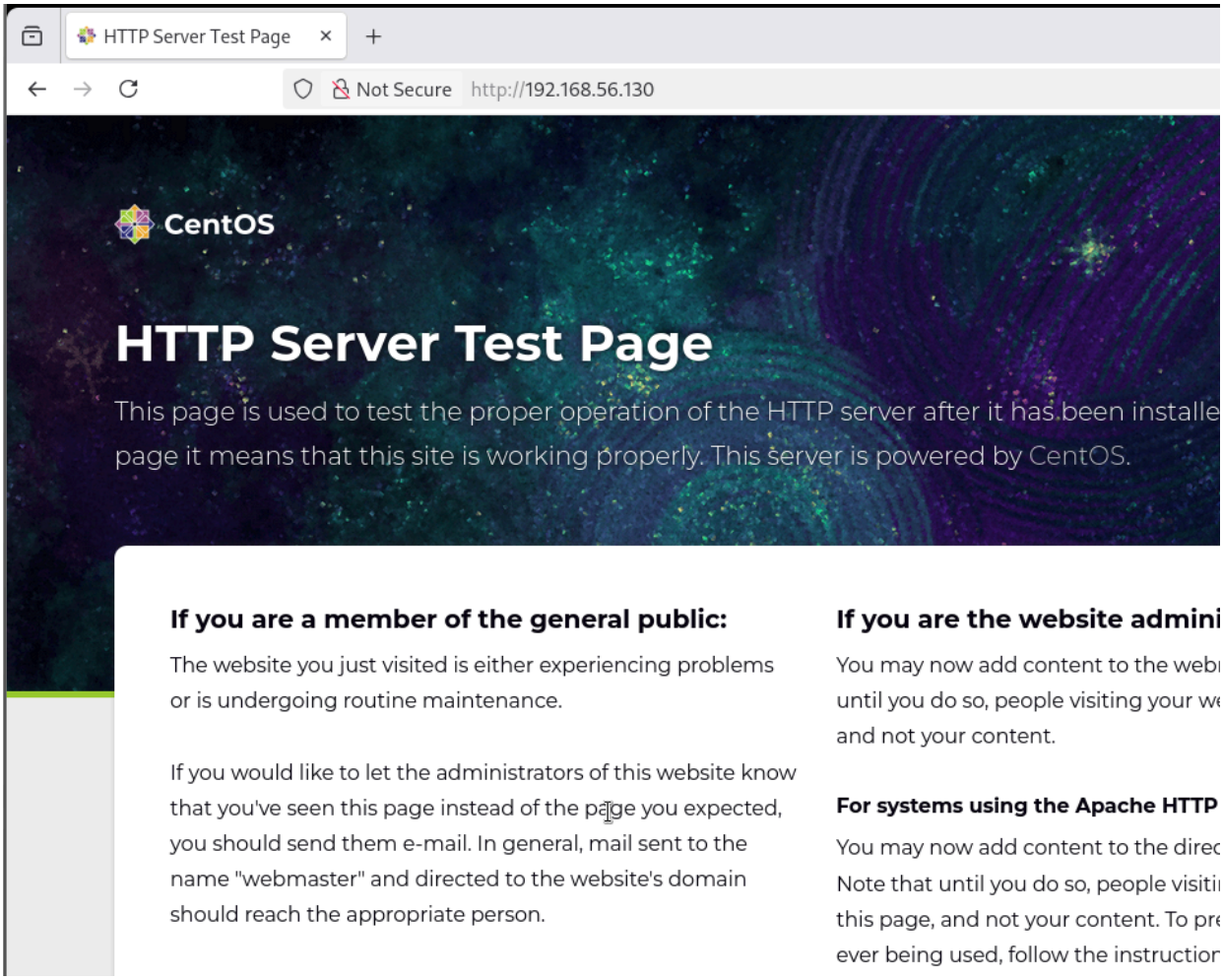
- name: start httpd (CentOS)
  tags: apache, centos,httpd
  services:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

2.


Activities
Firefox
Sep 3 10:05
Problem loading page
192.168.56.130
Unable to connect
Firefox can't establish a connection to the server at 192.168.56.130.
The site could be temporarily unavailable or too busy. Try again in a few moments.
If you are unable to load any pages, check your computer's network connection.
If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to a web.
Try

3.


TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.128]
ok: [192.168.56.130]
TASK [start httpd (CentOS)] ***************************
skipping: [192.168.56.128]
changed: [192.168.56.130]



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

In the supplementary activity I got rid of the upgrade in site.yml because it just took way too long to process and our internet connection is quite poor.

Supplementary Activity:
1. Creating a Role for Web Servers

```
roles/webserver:
defaults  files  handlers  meta  README.md  tasks  templates  tests  vars

roles/webserver/defaults:
main.yml
```

```
  GNU nano 7.2                roles/webserver/tasks/main.yml
---
- name: install apache and php for Ubuntu servers
  tags: apache, apache2, ubuntu
  apt:
    name: [apache2, libapache2-mod-php]
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name: [httpd, php]
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
```

Ansible-playbook -i inventory.ini site.yml –tags web –ask-become-pass

```
TASK [webserver : start httpd (CentOS)] ************************************
skipping: [192.168.56.103]
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [webserver : start apache2 (Ubuntu)] **********************************
skipping: [192.168.56.102]
ok: [192.168.56.103]
ok: [192.168.56.104]

PLAY [db_servers] **********************************************************

TASK [Gathering Facts] *****************************************************
ok: [192.168.56.103]
ok: [192.168.56.104]
ok: [192.168.56.102]

PLAY RECAP *****************************************************************
192.168.56.102             : ok=6    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.103             : ok=6    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.104             : ok=6    changed=0    unreachable=0    failed=0    s
```

2. Creating a Role for Database Servers

```
roles/database:
defaults  files  handlers  meta  README.md  tasks  templates  tests  vars

roles/database/defaults:
main.yml
```

```
  GNU nano 7.2                  roles/database/tasks/main.yml
---
- name: install mariadb package (CentOS)
  tags: centos, db, mariadb
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: install mariadb package (Ubuntu)
  tags: db, mariadb, ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: "Mariadb Restart/Enabling"
  tags: db, mariadb
  service:
    name: mariadb
    state: restarted
                              [ Read 21 lines ]
```

Ansible-playbook -i inventory.ini site.yml –tags db –ask-become-pass

```
TASK [database : install mariadb package (CentOS)] *****************************
skipping: [192.168.56.103]
skipping: [192.168.56.104]
changed: [192.168.56.102]

TASK [database : install mariadb package (Ubuntu)] *****************************
skipping: [192.168.56.102]
changed: [192.168.56.103]
changed: [192.168.56.104]

TASK [database : Mariadb Restart/Enabling] *************************************
changed: [192.168.56.104]
changed: [192.168.56.103]
changed: [192.168.56.102]

PLAY RECAP *********************************************************************
192.168.56.102             : ok=6    changed=2    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.103             : ok=6    changed=2    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.104             : ok=6    changed=2    unreachable=0    failed=0    s
```

3. Combining Multiple Roles in a Playbook

```
  GNU nano 7.2                          site.yml
---
- hosts: all
  become: true
  pre_tasks:

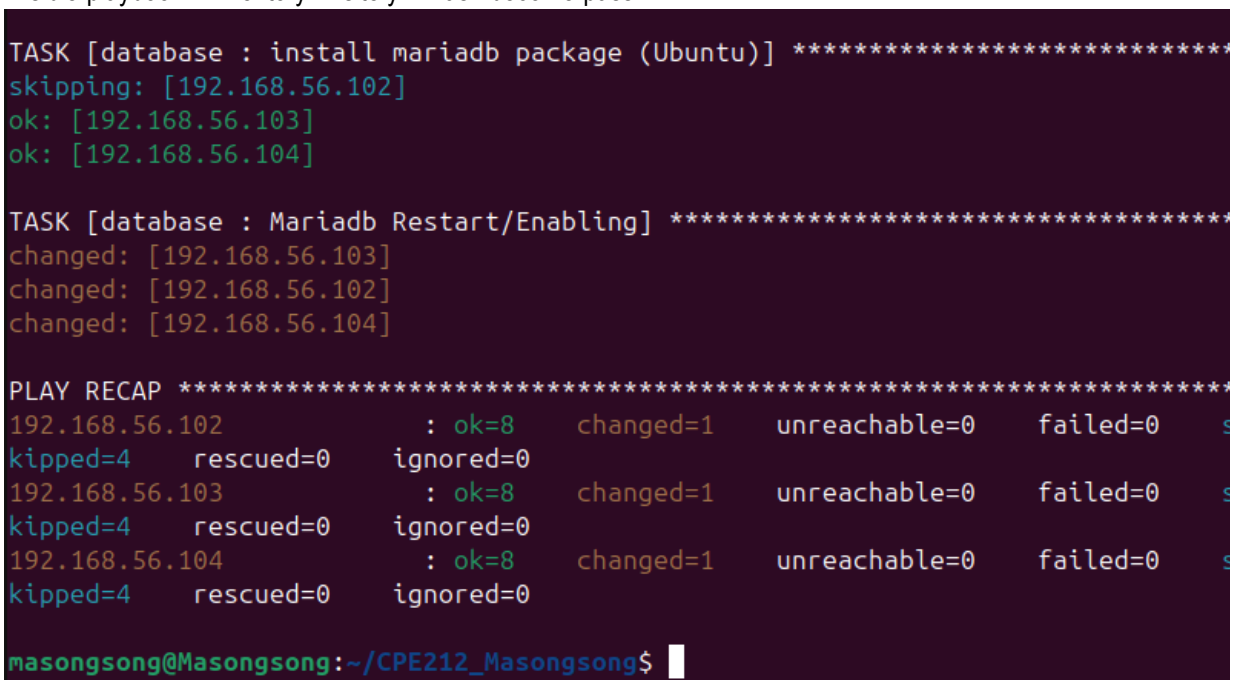
    - name: install updates (CentOS)
      tags: always
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        cache_valid_time: 3600
      when: ansible_distribution == "Ubuntu"
```

```
- hosts: web_servers
  become: true
  roles:
    - role: webserver
      tags: web

- hosts: db_servers
  become: true
  roles:
    - role: database
      tags: db
```

Ansible-playbook -i inventory.ini site.yml –ask-become-pass



Reflections:
Answer the following:

1. What is the importance of putting our remote servers into groups?
   - It makes it easier for us to implement it in our playbooks and can even possibly be reused in other projects.

2. What is the importance of tags in playbooks?
   - It specifies the play you want to run in the playbook, you can use it to filter out which plays you want to run in the playbook.

3. Why do you think some services need to be managed automatically in playbooks?
   - In order to reduce redundancy specifically those services that are a hassle to download via package installer in the OS.

Conclusion:
I have learned a lot from this HOA, mainly how to target specific nodes/plays in the playbook by tagging them and creating roles for them. Learning this knowledge allowed us to play specific nodes in the playbooks which lessens load times and reduces clutter in our said playbooks. We are also able to create roles with certain nicknames in order to make it easier to type rather than typing the whole keyword in order for the playbook to understand which to play. All in all I had fun and I definitely understood what I did and I hope to use this knowledge further down the line.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **NA**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NA**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **NA**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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